

Lecture 2: Brownian Motion

Lecturer: Andrew Papanicolaou

Scribe: Daniel Chertock

Note: *LaTeX template courtesy of UC Berkeley EECS department.*

Disclaimer: *This note has not been subjected to formal publication review. Distribution outside class requires instructor permission.*

2.1 Properties and Construction

Brownian motion is a continuous-time stochastic process with normal increments and variance scaled like time. An early application of this process in finance was the thesis of Louis Bachelier in the year 1900, “Théorie de la spéculation”. Bachelier to this day is still credited with the innovation of using stochastic processes to model asset prices.

On a probability space $(\Omega, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, the stochastic process $(W_t)_{t \geq 0}$ is **standard Brownian motion** with respect to $\mathcal{F}_t$ if it has the following four properties,

1. $W_0 = 0$,
2. For $0 \leq s < t$, $W_t - W_s$ is independent of $\mathcal{F}_s$ (i.e., increments are independent of the past),
3. Conditionally Gaussian: for $0 \leq s < t$, $W_t - W_s \sim \text{normal}(0, t - s)$,
4. With probability 1, the function $t \mapsto W_t$ is continuous.

Fact 1: W_t has the martingale property: for $t, s \geq 0$,

$$\begin{aligned}
 & \mathbb{E} \left[W_{t+s} \middle| \mathcal{F}_t \right] \\
 &= \mathbb{E} \left[W_{t+s} - W_t \middle| \mathcal{F}_t \right] + W_t \\
 &= \mathbb{E} [W_{t+s} - W_t] + W_t && \text{by increment's independence from past} \\
 &= W_t && \text{by mean zero of increments .}
 \end{aligned}$$

Fact 2: W_t has the Markov property: for $t, s \geq 0$,

$$\begin{aligned}
 & \mathbb{P}(W_{t+s} \leq w | \mathcal{F}_t) \\
 &= \mathbb{P}(W_{t+s} - W_t \leq w - W_t | \mathcal{F}_t) \\
 &= \mathbb{P}(W_{t+s} - W_t \leq w - W_t | W_t) && \text{by increment's independence from past} \\
 &= \mathbb{P}(W_{t+s} \leq w | W_t) .
 \end{aligned}$$

In general, stochastic processes with independent increments are Markovian.

2.1.1 Quadratic Variation

or any stochastic process, we measure quadratic variation to gauge the amount of movement a process makes over time. Consider the following partition of the time interval $[0, T]$:

$$0 = t_0 < t_1 < t_2 < t_3 < t_4 < \cdots < t_{n-1} < t_n = T .$$

Denote $h_n = \max_i(t_{i+1} - t_i)$, $n = 1, 2, \dots$, and let us consider sequences of partitions for which $h_n \rightarrow 0$ as $n \rightarrow \infty$. For a continuous-time stochastic process W_t , the quadratic variation is the limit of the sum of squared increments,

$$Q^n = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} [\Delta W_i]^2 ,$$

where $\Delta W_i = W_{t_{i+1}} - W_{t_i}$, with the limit holding in probability. It can be shown that the limit of Q^n as $h_n \rightarrow 0$ is T . Indeed, $\mathbb{E}Q^n = \sum_{i=0}^{n-1} \mathbb{E}\Delta W_i^2 = \sum_{i=0}^{n-1} (t_{i+1} - t_i) = T$, and

$$\begin{aligned} \text{var}(Q^n) &= \text{var}\left(\sum_{i=0}^{n-1} \Delta W_i^2\right) \\ &= \sum_{i=0}^{n-1} \text{var}(\Delta W_i^2) && \text{by independence of increments} \\ &= \sum_{i=0}^{n-1} \mathbb{E}\left(\Delta W_i^2 - (t_{i+1} - t_i)\right)^2 \\ &= \sum_{i=0}^{n-1} \left(\mathbb{E}\Delta W_i^4 - 2(t_{i+1} - t_i)\mathbb{E}\Delta W_i^2 + (t_{i+1} - t_i)^2\right) \\ &= 2 \sum_{i=0}^{n-1} (t_{i+1} - t_i)^2 \\ &\leq 2Th_n \\ &\rightarrow 0 , \end{aligned}$$

as $n \rightarrow \infty$. By Chebyshev's inequality, for any $\varepsilon > 0$,

$$\mathbb{P}(|Q^n - T| \geq \varepsilon) \leq \frac{\text{var}(Q^n)}{\varepsilon^2} \leq \frac{2T}{\varepsilon^2} h_n \rightarrow 0 ,$$

as $n \rightarrow \infty$. Therefore,

$$Q^n \rightarrow T \quad \text{in probability,}$$

and hence, the **quadratic variation** of a standard Brownian Motion over period $[0, T]$ is T .

W.P.1 Convergence (Optional Reading). For a given realization of the process, $\lim_{n \rightarrow \infty} Q^n \neq T$ if and only if $\exists \epsilon > 0$ such that $|Q^n - T| \geq \epsilon$ infinitely often (i.e., for infinitely-many n). Define the events $A_n := \{|Q^n - T| \geq \epsilon\}$. The Borel-Cantelli lemma says $\mathbb{P}(A_n \text{ i.o.}) = 0$ if $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$. We've already shown $\mathbb{P}(A_n) \leq 2Th_n/\epsilon^2$. Clearly, if $\sum_{n=1}^{\infty} h_n < \infty$ then $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$ for any $\epsilon > 0$, and therefore by Borel-Cantelli lemma it follows that $Q^n \rightarrow T$ w.p.1.

The above usage of Borel-Cantelli does not cover the case of the equally-spaced partition with $h_n = T/n$. In fact, equally-spaced partitions have strong convergence, but showing it requires a faster convergence in probability than Chebyshev has given. Consider a Markov inequality on the 4th moment,

$$\mathbb{P}(A_n) = \mathbb{P}\left((Q^n - T)^4 \geq \epsilon^4\right) \leq \frac{1}{\epsilon^4} \mathbb{E}(Q^n - T)^4 .$$

For $t_{i+1} - t_i \equiv T/n$ for all i , denote $X_i = \Delta W_i^2 - T/n$, and notice that $\mathbb{E}(Q^n - T)^4 = \sum_{i,j,k,\ell} \mathbb{E}X_i X_j X_k X_\ell = n\mathbb{E}X_i^4 + 3n(n-1)(\mathbb{E}X_i^2)^2$. For these moments, we have $\mathbb{E}X_i^2 = \mathbb{E}\Delta W_i^4 - (\mathbb{E}\Delta W_i^2)^2 = 2(T/n)^2$, and $\mathbb{E}X_i^4 = \mathbb{E}\Delta W_i^8 - 4(T/n)\mathbb{E}\Delta W_i^6 + 6(T/n)^2\mathbb{E}\Delta W_i^4 - 4(T/n)^3\mathbb{E}\Delta W_i^2 + (T/n)^4 = (105 - 60 + 18 - 4 + 1)(T/n)^4 = 60(T/n)^4$. Therefore, $\mathbb{P}(A_n) \leq \frac{T^4}{(\epsilon n)^4}(60n + 6n(n-1)) \leq \frac{C}{\epsilon^4 n^2}$, and thus $\sum_n \mathbb{P}(A_n) \leq \frac{C}{\epsilon^4} \sum_n \frac{1}{n^2} < \infty$, and by Borel-Cantelli lemma, $Q^n \rightarrow T$ w.p.1 for the sequence of equally-spaced partitions.

In general, the limit of quadratic variation w.p.1 holds for any partitions with $h_n = o\left(\frac{1}{\log n}\right)$.

2.1.2 Random Walks

Simplest way to think of Brownian motion is as a sum of Bernoulli variables. Let Z_i be the result of a draw from a Bernoulli at time t so that $\mathbb{P}(Z_i = 1) = \mathbb{P}(Z_i = -1) = 0.5$. An interval $[0, T]$ can be split into sub-intervals of length $\Delta t = T/n$. We can represent one-step of a random walk as

$$W_t^n = \sqrt{T} \left(\frac{1}{\sqrt{n}} \sum_{i=0}^{\lfloor nt/T \rfloor} Z_i \right) .$$

Clearly we have $\mathbb{E}W_t^n = 0$ and $\text{var}(W_t^n) = \frac{T}{n} \lfloor nt/T \rfloor \rightarrow t$ as $n \rightarrow \infty$. In addition, by the Central Limit Theorem,

$$\sqrt{\frac{T}{n}} \sum_{i=0}^{\lfloor nt/T \rfloor} Z_i \Rightarrow \text{normal}(0, t) ,$$

as $n \rightarrow \infty$. It can be shown that the path of process $(W_t^n)_{t \in [0, T]}$ converges in distribution to a continuous process that satisfies the four properties of Brownian motion. Figure 2.1 shows paths of this discrete random walk over $[0, 1]$ with $n = 1000$, with the sideways parabola being $\pm 1.96 \times \sqrt{t}$ (i.e., the 95% confidence band for the process). In Figure 2.2 we see a histogram of this discrete random walk at time $t = 1$ with a standard normal density laid over. It is clear from the plot that the simulated paths are approximately normally distributed, as predicted by the Central Limit Theorem.

2.1.3 Arithmetic Brownian Motion

A process X_t with increments having non-zero drift and non-unit volatility is

$$X_t - X_0 = \mu t + \sigma W_t .$$

We call this process an **arithmetic Brownian motion**. It has a drift of μ and a variance of σ^2 per unit time. The process X_t has normally distributed increments,

$$X_{t+s} - X_t \sim \text{normal}(\mu s, \sigma^2 s) ,$$

and it can be shown that the quadratic variation on $[0, t]$ is $\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} \Delta X_i^2 = \sigma^2 t$.

2.2 Examples

Example 1 (Reflection Principle). *For scalar $a > 0$, define a stopping time $\tau = \inf\{t > 0 | W_t \geq a\}$, which is the first time that W_t is above a . The distribution of τ can be calculated using the symmetry of the W_t 's*

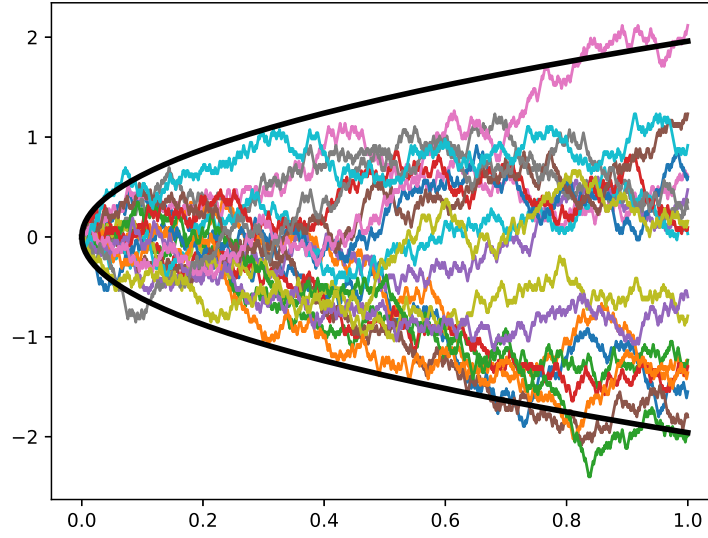


Figure 2.1: A discrete random walk over $[0,1]$ with $n = 1000$. The sideways parabola is $\pm 1.96 \times \sqrt{t}$, which shows the 95% confidence band for the process.

distribution at zero,

$$\begin{aligned}
 \mathbb{P}(W_t > a) &= \mathbb{P}(W_t > a | \tau \leq t) \mathbb{P}(\tau \leq t) + \underbrace{\mathbb{P}(W_t > a | \tau > t)}_{=0} \mathbb{P}(\tau > t) \\
 &= \mathbb{P}(W_t - W_\tau > 0 | \tau \leq t) \mathbb{P}(\tau \leq t) \\
 &= \frac{1}{2} \mathbb{P}(\tau \leq t) ,
 \end{aligned}$$

where we have used the fact that $\mathbb{P}(W_\tau = a) = 1$ by continuity of W_t . Therefore,

$$\mathbb{P}(\tau \leq t) = 2\mathbb{P}(W_t > a) = 2(1 - \Phi(a/\sqrt{t})) = 2\Phi(-a/\sqrt{t}) ,$$

where Φ is the standard normal cumulative distribution function. Notice that this give us the distribution of $M_t = \max_{s \leq t} W_s$,

$$\mathbb{P}(M_t \leq a) = \mathbb{P}(\tau \geq t) = 1 - 2\Phi(-a/\sqrt{t}) ,$$

for any $a \geq 0$.

Example 2 (Probability of Ruin). For scalars $a, b > 0$, define a process

$$X_t = X_0 + W_t ,$$

with $-b \leq X_0 \leq a$, and define the stopping time $\tau = \inf\{t > 0 | W_t \in \{-b, a\}\}$. If $W_\tau = -b$ then the process has reached ‘ruin’, and if $W_\tau = a$ then the process ends in ‘success’. Using the martingale property of X_t ,

$$X_0 = \mathbb{E}X_\tau = -b\mathbb{P}(X_\tau = -b) + a(1 - \mathbb{P}(X_\tau = -b)) ,$$

and so the probability of ruin is

$$\mathbb{P}(X_\tau = -b) = \frac{a - X_0}{a + b} .$$

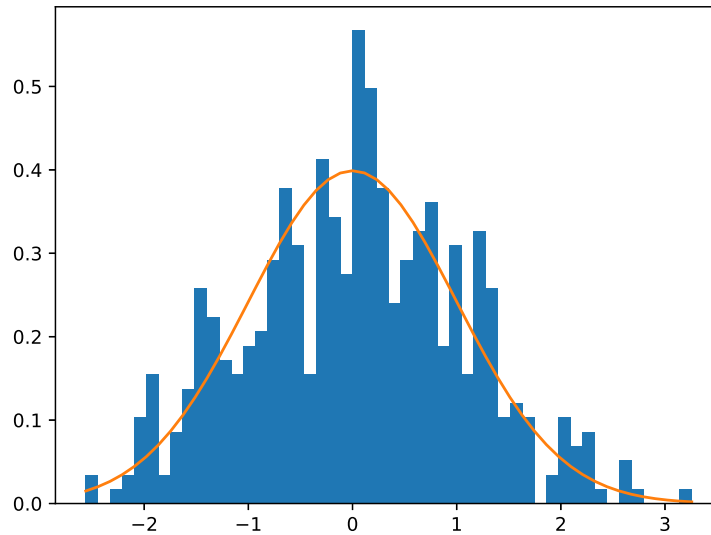


Figure 2.2: The histogram of the discrete random walk at time $t = 1$, with a standard normal density laid over. Clearly the simulated paths are approximately normally distributed, as predicted by the Central Limit Theorem.

2.3 Remarks

1. The study of Brownian motion is deep and is worthy of thorough investigation—both for industry practitioners and academics. A good reference for Brownian motion is [1] and [3].
2. A good background reference for probability and stochastic processes is [2].

References

- [1] T. Björk. *Arbitrage Theory in Continuous Time*. Oxford University Press, Oxford, UK, 4th edition, 2020.
- [2] R. Durrett. *Probability: Theory and Examples*, volume 49. Cambridge University Press, 2019.
- [3] B. Oksendal. *Stochastic differential equations: an introduction with applications*. Springer Science & Business Media, 2013.